

Operating Systems Lab All Tasks

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Section :B

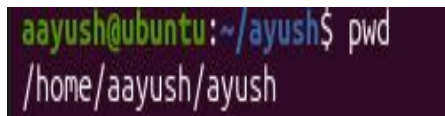
Task 1 – Display the current working directory

Command :

`pwd`

Explanation: Displays the complete path of the current working directory.

Output/Screenshot:



```
aayush@ubuntu:~/ayush$ pwd
/home/aayush/ayush
```

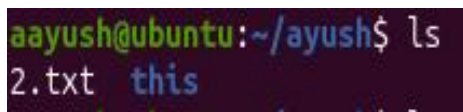
Task 2 – Display the contents of the current directory

Command :

`ls`

Explanation: Lists files and directories in the current directory.

Output/Screenshot:



```
aayush@ubuntu:~/ayush$ ls
2.txt  this
```

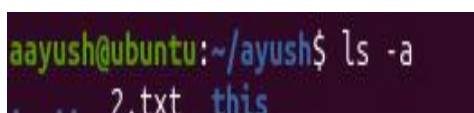
Task 3 – Display all files, including hidden files

Command :

`ls -a`

Explanation: Displays all files and directories, including hidden files.

Output/Screenshot:



```
aayush@ubuntu:~/ayush$ ls -a
.  ..  2.txt  this
```

Task 4 – Create Linux_Assignment

Command:

```
mkdir Linux_Assignment
```

Explanation: Creates the Linux_Assignment directory.

Output/Screenshot:



```
aayush@ubuntu:~/ayush$ mkdir Linux_Assignment
```

Task 5 – Enter Linux_Assignment

Command:

```
cd Linux_Assignment
```

Explanation: Changes to the Linux_Assignment directory.

Output/Screenshot:



```
aayush@ubuntu:~/Desktop$ cd Linux_Assignment
```

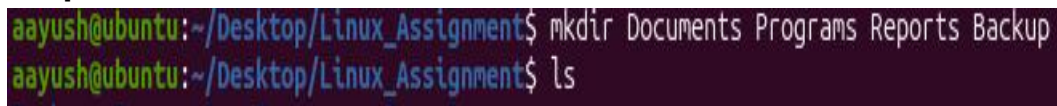
Task 6 – Create four directories

Command:

```
mkdir Documents Programs Reports Backup
```

Explanation: Creates the four required directories.

Output/Screenshot:



```
aayush@ubuntu:~/Desktop/Linux_Assignment$ mkdir Documents Programs Reports Backup  
aayush@ubuntu:~/Desktop/Linux_Assignment$ ls
```

Task 7 – Verify the directories

Command:

```
ls
```

Explanation: Displays and verifies Documents, Programs, Reports, and Backup.

Output/Screenshot:



```
aayush@ubuntu:~/Desktop/Linux_Assignment$ ls  
Backup Documents Programs Reports
```

Task 8 – Create three files

Command:
cd Documents
touch file1.txt file2.txt file3.txt

Explanation: Enters Documents and creates the three text files.

Output/Screenshot:

```
ayush@ubuntu:~/Desktop/Linux_Assignment$ cd Documents
ayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ touch file1.txt file2.txt file3.txt
```

Task 9 – Display the files

Command:
ls

Explanation: Lists the files in Documents.

Output/Screenshot:

```
ayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ ls
file1.txt file2.txt file3.txt
```

Task 10 – Write text into file1.txt

Command:
echo "Operating Systems" > file1.txt
echo "Linux Command Practice" >> file1.txt
echo "KRMU" >> file1.txt

Explanation: Writes the required text into file1.txt.

Output/Screenshot:

```
ayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ echo "Linux Command Practice" >> file1.txt
ayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ echo "Krmu" >> file1.txt
```

Task 11 – Display file1.txt

Command:
cat file1.txt

Explanation: Displays the contents of file1.txt.

Output/Screenshot:

```
ayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ cat file1.txt
Linux Command Practice
```

Task 12 – Copy file1.txt to Backup

Command:
cp file1.txt ../Backup/

Explanation: Copies file1.txt to Backup.

Output/Screenshot:

```
ayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ cp file1.txt ../Backup/
```

Task 13 – Rename file2.txt

Command:

```
mv file2.txt student.txt
```

Explanation: Renames file2.txt to student.txt.

Output/Screenshot:

```
ayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ mv file2.txt student.txt
```

Task 14 – Move file3.txt to Reports

Command:

```
mv file3.txt ../Reports/
```

Explanation: Moves file3.txt to Reports.

Output/Screenshot:

```
ayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ mv file3.txt ../Report/
mv: cannot move 'file3.txt' to '../Report/': Not a directory
```

Task 15 – Verify Backup

Command:

```
ls ../Backup
```

Explanation: Displays Backup and verifies file1.txt.

Output/Screenshot:

```
ayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ ls ../Backup
file1.txt
```

Task 16 – Count lines, words and characters

Command:

```
wc file1.txt
```

Explanation: Displays the counts for file1.txt.

Output/Screenshot:

```
ayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ wc file1.txt
1  3 23 file1.txt
```

Task 17 – First few lines

Command:
head file1.txt

Explanation: Displays the beginning of file1.txt.

Output/Screenshot:

```
ayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ head file1.txt
linux Command Practice
```

Task 18 – Last few lines

Command:
tail file1.txt

Explanation: Displays the end of file1.txt.

Output/Screenshot:

```
ayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ tail file1.txt
linux Command Practice
```

Task 19 – Search for Linux

Command:
grep "Linux" file1.txt

Explanation: Searches for the word Linux in file1.txt.

Output/Screenshot:

```
ayush@ubuntu:~$ grep "Linux" file1.txt
```

Task 20 – Find all .txt files

Command:
cd ..
find . -name "*.txt"

Explanation: Finds all .txt files under Linux_Assignment.

Output/Screenshot:

```
ayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ cd ..
ayush@ubuntu:~/Desktop/Linux_Assignment$ find . -name "*.txt"
./Reports/file3.txt
./Backup/file1.txt
./Documents/student.txt
./Documents/file1.txt
```

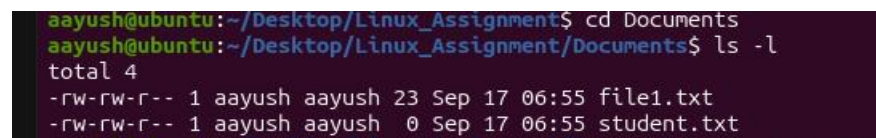
Task 21 – Display permissions

Command:

```
cd Documents  
ls -l
```

Explanation: Displays detailed file permissions.

Output/Screenshot:



```
aayush@ubuntu:~/Desktop/Linux_Assignment$ cd Documents  
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ ls -l  
total 4  
-rw-rw-r-- 1 aayush aayush 23 Sep 17 06:55 file1.txt  
-rw-rw-r-- 1 aayush aayush  0 Sep 17 06:55 student.txt
```

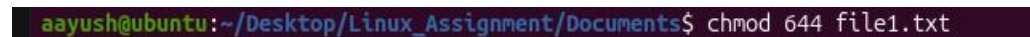
Task 22 – Change permissions

Command:

```
chmod 644 file1.txt
```

Explanation: Owner gets read/write; group and others get read-only.

Output/Screenshot:



```
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ chmod 644 file1.txt
```

Task 23 – Verify permissions

Command:

```
ls -l file1.txt
```

Explanation: Verifies the permission setting.

Output/Screenshot:



```
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ ls -l file1.txt  
file1.txt
```

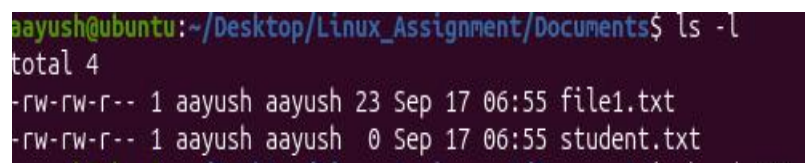
Task 24 – Explain -rwxr-xr--

Command:

```
- rwx r-x r--  
  Owner Group Others  
Numeric form: 754
```

Explanation: Regular file; owner has rwx, group has r-x, others have r--.

Output/Screenshot:



```
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ ls -l  
total 4  
-rw-rw-r-- 1 aayush aayush 23 Sep 17 06:55 file1.txt  
-rw-rw-r-- 1 aayush aayush  0 Sep 17 06:55 student.txt
```

Task 25 – uname -a

Command:
uname -a

Explanation: Displays detailed Linux system and kernel information.

Output/Screenshot:

```
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ uname -a
Linux ubuntu 7.0.0-31-generic #31-Ubuntu SMP PREEMPT_DYNAMIC Sat Aug 1 04:26:38 UTC 2026 x86_64 GNU/Linux
```

Task 26 – whoami

Command:
whoami

Explanation: Displays the current username.

Output/Screenshot:

```
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ whoami
aayush
```

Task 27 – date

Command:
date

Explanation: Displays the current date and time.

Output/Screenshot:

```
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ date
Thu Sep 17 07:08:37 AM UTC 2026
```

Task 28 – df -h

Command:
df -h

Explanation: Displays disk usage in human-readable form.

Output/Screenshot:

```
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           1.6G  3.3M  1.6G   1% /run
/dev/sda2       18G   6.8G  9.6G  42% /
tmpfs           4.0G   0  4.0G   0% /dev/shm
tmpfs           4.0G  32K  4.0G   1% /tmp
none            1.0M   0  1.0M   0% /run/credentials/systemd-journald.service
none            1.0M   0  1.0M   0% /run/credentials/systemd-resolved.service
tmpfs           803M 128K  803M   1% /run/user/1000
```

Task 29 – free -h

Command:
free -h

Explanation: Displays RAM and swap usage.

Output/Screenshot:

```
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ free -h
              total        used        free      shared  buff/cache   available
Mem:          7.8Gi        1.5Gi        4.6Gi         50Mi        2.1Gi        6.3Gi
Swap:          0B           0B           0B
```

Task 30 – uptime

Command:
uptime

Explanation: Displays system uptime and load information.

Output/Screenshot:

```
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ uptime
07:08:57 up 58 min,  1 user,  load average: 1.05, 0.53, 0.39
```

Task 31 – Display running processes

Command:
ps

Explanation: Displays processes associated with the current terminal.

Output/Screenshot:

```
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ ps
  PID TTY          TIME CMD
 14714 pts/1        00:00:00 bash
 15391 pts/1        00:00:00 ps
```

Task 32 – Detailed processes

Command:
ps aux

Explanation: Displays detailed information about system processes.

Output/Screenshot:

```
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1   0.5  0.2 26572 17428 ?        Ss   06:10   0:20 /usr/lib/systemd/systemd --switched-roo
root         2   0.0  0.0      0     0 ?        S    06:10   0:00 [kthreadd]
root         3   0.0  0.0      0     0 ?        S    06:10   0:00 [pool_workqueue_release]
root         4   0.0  0.0      0     0 ?        I<   06:10   0:00 [kworker/R-rcu_gp]
root         5   0.0  0.0      0     0 ?        I<   06:10   0:00 [kworker/R-sync_wq]
root         6   0.0  0.0      0     0 ?        I<   06:10   0:00 [kworker/R-kvfree_rcu_reclaim]
root         7   0.0  0.0      0     0 ?        I<   06:10   0:00 [kworker/R-slub_flushwq]
root         8   0.0  0.0      0     0 ?        I<   06:10   0:00 [kworker/R-netns]
root        12   0.0  0.0      0     0 ?        I    06:10   0:00 [kworker/u28:0-events_unbound]
root        13   0.0  0.0      0     0 ?        I<   06:10   0:00 [kworker/R-mm_percpu_wq]
root        14   0.0  0.0      0     0 ?        S    06:10   0:01 [ksoftirqd/0]
root        15   0.2  0.0      0     0 ?        I    06:10   0:08 [rcu_preempt]
root        16   0.0  0.0      0     0 ?        S    06:10   0:00 [rcu_exp_par_gp_kthread_worker/0]
root        17   0.1  0.0      0     0 ?        S    06:10   0:04 [rcu_exp_gp_kthread_worker]
root        18   0.0  0.0      0     0 ?        S    06:10   0:00 [migration/0]
root        19   0.0  0.0      0     0 ?        S    06:10   0:00 [kprobe-optimizer]
root        20   0.0  0.0      0     0 ?        S    06:10   0:00 [idle_inject/0]
root        21   0.0  0.0      0     0 ?        S    06:10   0:00 [cpuhp/0]
root        22   0.0  0.0      0     0 ?        S    06:10   0:00 [cpuhp/1]
root        23   0.0  0.0      0     0 ?        S    06:10   0:00 [idle_inject/1]
root        24   0.0  0.0      0     0 ?        S    06:10   0:00 [migration/1]
root        25   0.0  0.0      0     0 ?        S    06:10   0:00 [ksoftirqd/1]
root        28   0.0  0.0      0     0 ?        S    06:10   0:00 [cpuhp/2]
root        29   0.0  0.0      0     0 ?        S    06:10   0:00 [idle_inject/2]
root        30   0.0  0.0      0     0 ?        S    06:10   0:00 [migration/2]
root        31   0.0  0.0      0     0 ?        S    06:10   0:02 [ksoftirqd/2]
root        34   0.0  0.0      0     0 ?        S    06:10   0:00 [cpuhp/3]
root        35   0.0  0.0      0     0 ?        S    06:10   0:00 [idle_inject/3]
root        36   0.0  0.0      0     0 ?        S    06:10   0:00 [migration/3]
root        37   0.0  0.0      0     0 ?        S    06:10   0:01 [ksoftirqd/3]
root        39   0.0  0.0      0     0 ?        I<   06:10   0:00 [kworker/3:0H-kblockd]
root        40   0.0  0.0      0     0 ?        S    06:10   0:00 [cpuhp/4]
root        41   0.0  0.0      0     0 ?        S    06:10   0:00 [idle_inject/4]
root        42   0.0  0.0      0     0 ?        S    06:10   0:00 [migration/4]
root        43   0.0  0.0      0     0 ?        S    06:10   0:02 [ksoftirqd/4]
root        45   0.0  0.0      0     0 ?        I<   06:10   0:00 [kworker/4:0H-kblockd]
root        46   0.0  0.0      0     0 ?        S    06:10   0:00 [cpuhp/5]
root        47   0.0  0.0      0     0 ?        S    06:10   0:00 [idle_inject/5]
root        48   0.0  0.0      0     0 ?        S    06:10   0:00 [migration/5]
root        49   0.0  0.0      0     0 ?        S    06:10   0:00 [ksoftirqd/5]
```

Task 33 – Start a process

Command:
sleep 100

Explanation: Starts a process that sleeps for 100 seconds.

Output/Screenshot:

```
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ sleep 100 &
[1] 15410
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ pgrep sleep
15410
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ kill 15410
[1]+  Terminated                  sleep 100
```

Task 34 – Find PID

Command:
pgrep sleep

Explanation: Displays the PID of the sleep process.

Output/Screenshot:

```
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ pgrep sleep
15410
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ kill 15410
```

Task 35 – Terminate sleep process

Command:

kill <PID>

Explanation: Terminates the process using its PID; replace <PID> with the actual PID.

Output/Screenshot:

```
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ kill <PID>
```

Task 36 – Verify termination

Command:

pgrep sleep

Explanation: No output means the sleep process has terminated.

Output/Screenshot:

```
aayush@ubuntu:~$ pgrep sleep
```

Task 37 – Create students.txt

Command:

printf "Aman\nRiya\nRahul\nPriya\nNeha\nKaran\n" > students.txt

Explanation: Creates students.txt with the six required names.

Output/Screenshot:

```
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ printf "Aman\nRiya\nRahul\nPriya\nNeha\nKaran\n" > students.txt
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ cat students.txt
Aman
Riya
Rahul
Priya
Neha
Karan
```

Task 38 – Display students.txt

Command:

cat students.txt

Explanation: Displays all student names.

Output/Screenshot:

```
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ sleep 100 &
[1] 15410
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ pgrep sleep
15410
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ kill 15410
[1]+  Terminated                  sleep 100
```

Task 39 – Count students

Command:

```
wc -l students.txt
```

Explanation: Counts the six student-name lines.

Output/Screenshot:

```
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ wc -l students.txt
5 students.txt
```

Task 40 – Sort

alphabetically

Command:

```
sort students.txt
```

Explanation: Sorts the names alphabetically.

Output/Screenshot:

```
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ sort students.txt
Aman
Karan
Neha
Rahul
Riya
```

Task 41 – Reverse order

Command:

```
sort -r students.txt
```

Explanation: Displays names in reverse alphabetical order.

Output/Screenshot:

```
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ sort -r students.txt
Riya
Rahul
Neha
Karan
Aman
```

Task 42 – Use a pipe

Command:

```
cat students.txt | wc -l
```

Explanation: Sends cat output through a pipe to wc -l.

Output/Screenshot:

```
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ cat students.txt
Aman
Riya
Rahul
Neha
Karan
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ wc -l students.txt
5 students.txt
```

Task 43 – Create OS_Lab structure

Command:

```
cd ~
mkdir -p OS_Lab/Programs/C
mkdir -p OS_Lab/Programs/Python
mkdir -p OS_Lab/Assignments
mkdir -p OS_Lab/Backup
```

Explanation: Creates the required OS_Lab structure.

Output/Screenshot:

```
aayush@ubuntu:~/Desktop/Linux_Assignment/Documents$ cd ~
aayush@ubuntu:~$ mkdir -p OS_Lab/Programs/C
aayush@ubuntu:~$ mkdir -p OS_Lab/Assignments
aayush@ubuntu:~$ mkdir -p OS_Lab/Backup
aayush@ubuntu:~$ find OS_Lab -type d
OS_Lab
OS_Lab/Assignments
OS_Lab/Programs
OS_Lab/Programs/C
OS_Lab/Backup
```

Task 44 – Create Python files

Command:

```
cd ~/OS_Lab/Programs/Python
touch hello.py addition.py calculator.py
```

Explanation: Creates the three Python files.

Output/Screenshot:

```
aayush@ubuntu:~$ cd ~/OS_Lab/Programs/Python
bash: cd: /home/aayush/OS_Lab/Programs/Python: No such file or directory
aayush@ubuntu:~$ touch hello.py addition.py calculate.py
aayush@ubuntu:~$ ls
Desktop  Downloads  OS_Lab  Public  Videos  ayush  hello.py
Documents Music  Pictures Templates addition.py calculate.py snap
```

Task 45 – Write and run hello.py

Command:

```
printf 'print("Hello Linux")\nprint("Welcome to Operating Systems Lab")\n' >
```

```
hello.py
python3 hello.py
```

Explanation: Creates and runs the required Python program.

Output/Screenshot:

```
aayush@ubuntu:~$ printf 'print(Hello Linux)\nprint("Welcome to Operating Systems Lab")\n' >hello.py
aayush@ubuntu:~$ cat hello.py
print(Hello Linux)
print("Welcome to Operating Systems Lab")
```

Task 46 – Backup OS_Lab

Command:

```
cd ~
tar -czvf OS_Lab_Backup.tar.gz OS_Lab
```

Explanation: Creates a compressed backup archive of OS_Lab.

Output/Screenshot:

```
aayush@ubuntu:~$ cd ~
aayush@ubuntu:~$ tar -czvf OS_Lab_Backup.tar.gz OS_Lab
OS_Lab/
OS_Lab/Assignments/
OS_Lab/Programs/
OS_Lab/Programs/C/
OS_Lab/Backup/
```

Task 47 – Display OS_Lab size

Command:

```
du -sh OS_Lab
```

Explanation: Displays the total size of OS_Lab in human-readable format.

Output/Screenshot:

```
aayush@ubuntu:~$ du -sh OS_Lab
20K    OS_Lab
```

VIVA QUESTIONS – ANSWERS

1. What is a PID?

PID stands for Process ID. It is the numerical identifier assigned to a running process.

2. What is the purpose of ps?

ps displays information about running processes.

3. What is the purpose of kill?

kill sends a signal to a process, commonly requesting it to terminate.

4. What is a pipe (|)?

A pipe sends the output of one command as the input to another command.

5. What is input/output redirection?

It changes where a command gets input from or sends output to, such as a file.

6. Difference between > and >>?

> creates or overwrites a file, while >> appends content to the end of a file.

7. Why is python3 commonly used instead of python in Ubuntu?

python3 explicitly runs Python 3 and avoids ambiguity about which Python version is being used.